INTERMEDIATE MACROECONOMICS

IS-LM MODEL OF BUSINESS CYCLES

7. EXPENDITURE FUNCTION

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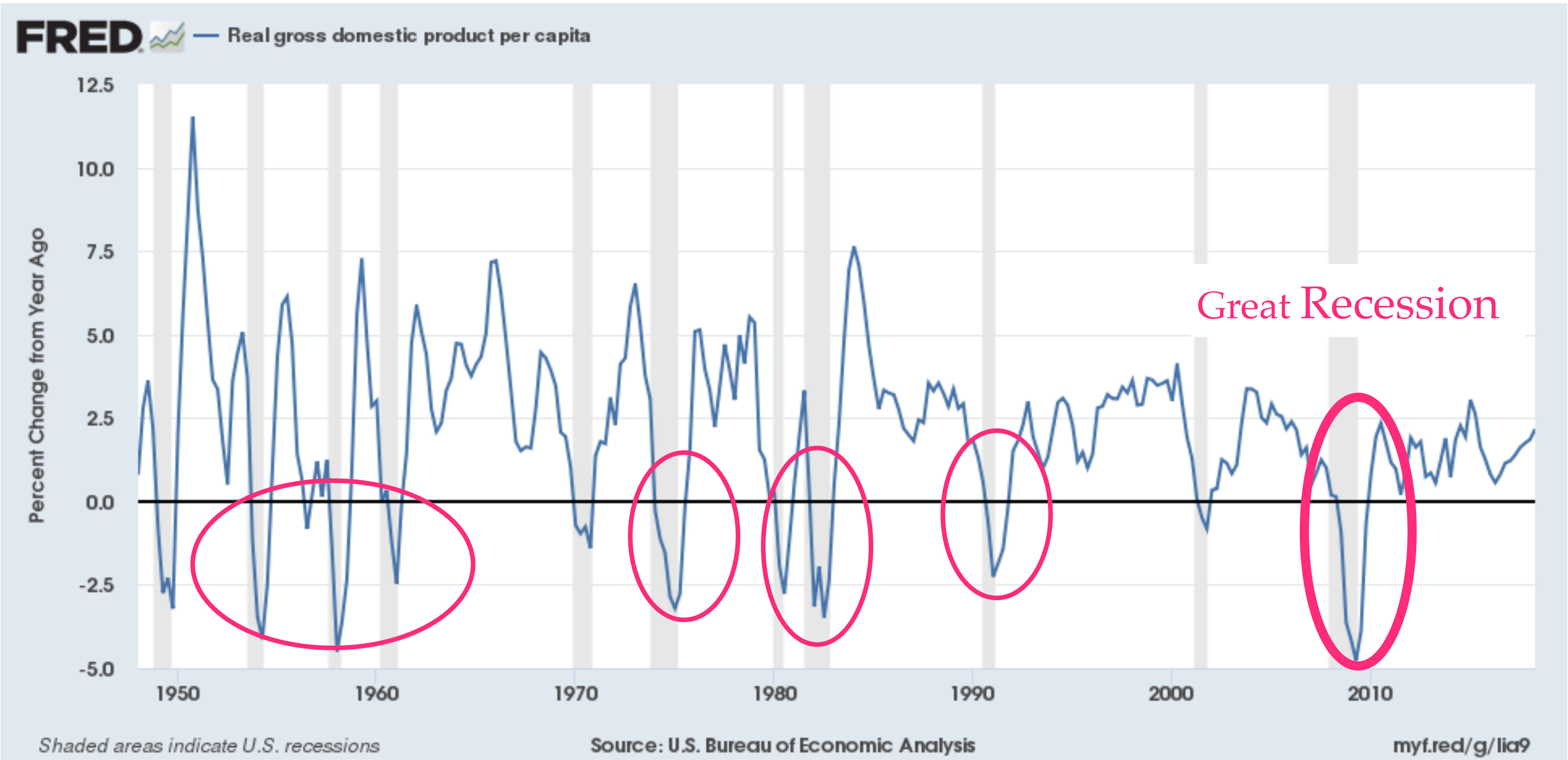
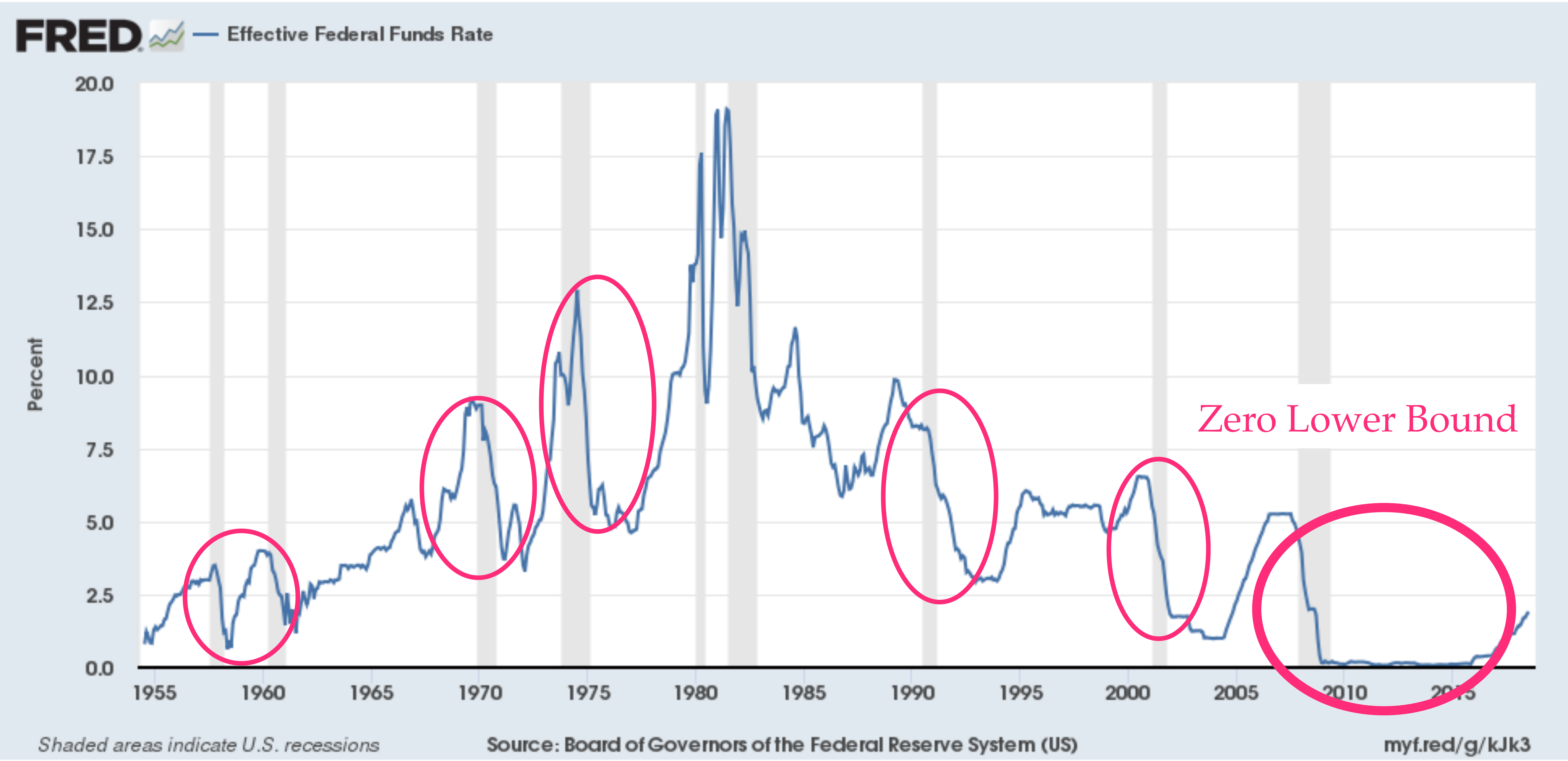
MOTIVATION FOR IS-LM MODEL: GROWTH RATE OF GDP/CAPITA



MOTIVATION FOR IS-LM MODEL: US MONETARY POLICY



EXPENDITURE ON GOODS (Z)

- expenditure on consumption goods + newly produced capital goods by
  - households
  - firms
  - government
- assumption: closed economy so exports = imports = 0
- expenditure on goods is $Z = C + G + I$

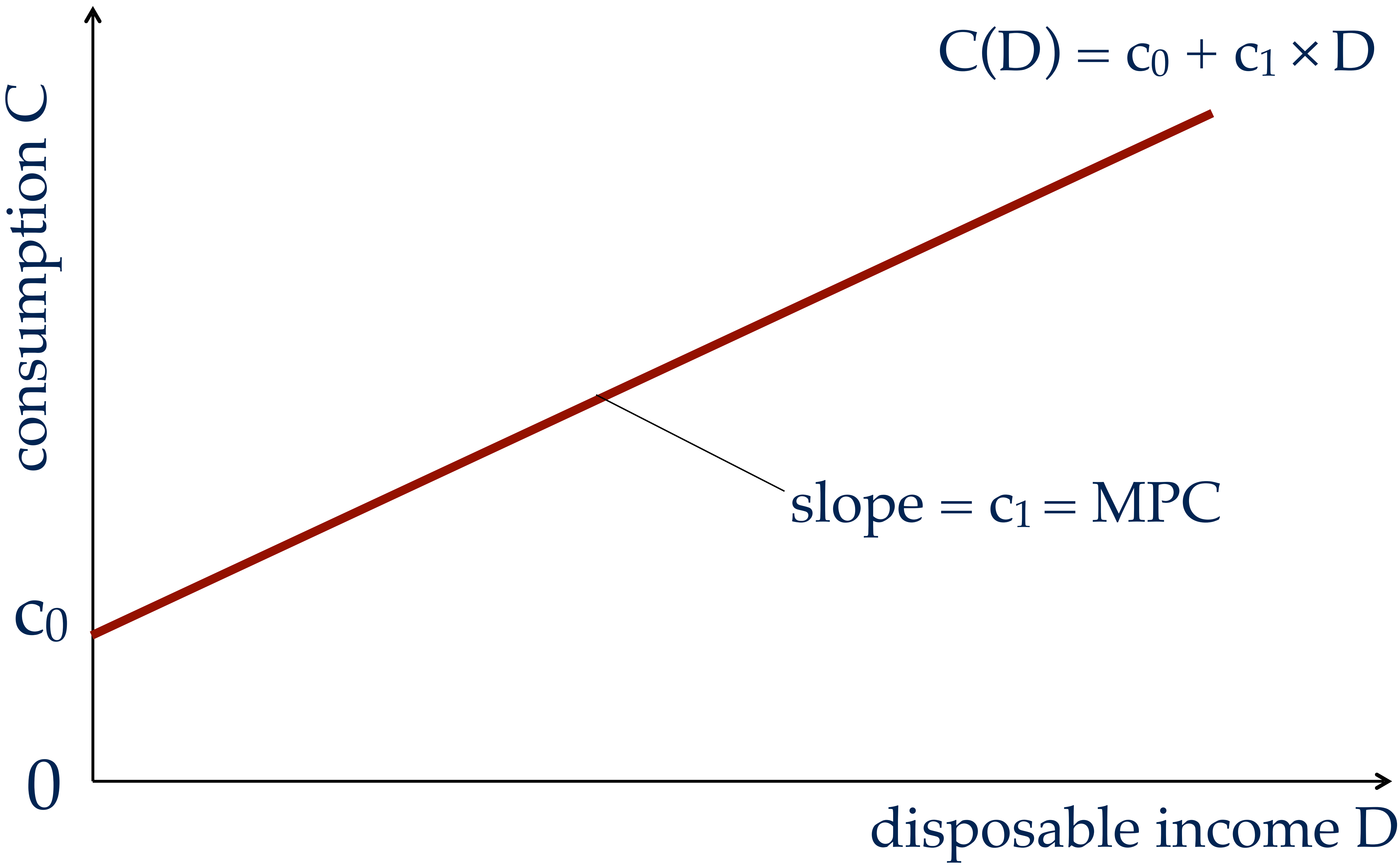
CONSUMPTION BY HOUSEHOLDS (C)

- consumption is an increasing function of disposable income (D)
    - $C = C(D)$ with $C'(D) > 0$
- shape of consumption function: $C(D) = c_0 + c_1 \times D$
    - $c_0 > 0$: what people consume if $D = 0$
        - to consume without disposable income, people use their savings
    - $0 < c_1 < 1$: the marginal propensity to consume (MPC)
        - when D increases by \$1, consumption increases by MPC
        - $0 < MPC$: people spend part of any increase in D
        - $MPC < 1$: people save part of any increase in D

DISPOSABLE INCOME (D)

- disposable income is the income remaining once households have paid their taxes & received transfers: $D = Y - T$
- Y = total income
- T = taxes – government transfers
  - taxes: income tax, property tax, sales tax, payroll tax
  - transfers: Social Security, Medicare, Medicaid, UI
  - $T > 0$: household pays more taxes than it receives benefits
  - $T < 0$: household receives more benefits than it pays taxes
- T is a parameter of the model (taken as given)

CONSUMPTION FUNCTION



OTHER COMPONENTS OF Z

- recall: $Z = C + I + G$
    - C is given by the consumption function, describing households' behavior
- government expenditure ($G > 0$): parameter of the model (taken as given)
- investment ($I > 0$): parameter of the model (for now)
    - in the complete IS-LM model, investment will be a function of the interest rate and aggregate income

TOTAL EXPENDITURE ON GOODS

- $Z = C + I + G + NX$
- $Z = C(Y–T) + I + G$ [NX=0 & C=C(D) & D=Y-T]
- $Z = c_0 + c_1 \times (Y – T) + I + G$ [using C(D)]
- $Z = [c_0 + I + G – c_1 \times T] + c_1 \times Y$ [algebra]

EXPENDITURE FUNCTION

- $Z(Y) = [c_0 + I + G - c_1 \times T] + c_1 \times Y$

- $[c_0 + I + G - c_1 \times T] > 0$: autonomous expenditure

  - with balanced government budget, $G = T$, and then $G - c_1 \times T = (1 - c_1) \times G > 0$; furthermore $c_0 > 0$ and $I > 0$; so autonomous spending > 0

  - autonomous expenditure is the amount of expenditure when income is 0

- $c_1 \times Y$: consumers' expenditure out of income